Machine Learning-Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Project Report | Unified Mentor / Parcl project brief

DATA DISCLAIMER: All numerical results in this report are generated from synthetic practice data, not actual Parcl customer records. Results are illustrative and must not be represented as company findings.

Abstract

This project demonstrates a machine-learning workflow for exploring real-estate buyer profiles using buyer type, geography, acquisition purpose, financing indicator, referral channel, age, and satisfaction. It includes data cleaning, feature engineering, encoding, scaling, EDA, K-Means, hierarchical clustering, model evaluation, and a Streamlit dashboard. The supplied dataset is synthetic.

1. Problem Statement and Objectives

The brief identifies limited data-driven understanding of buyer types, investment motivations, geographic differences, and financing patterns. Objectives include cleaning buyer data, engineering age, encoding and scaling features, comparing K-Means and hierarchical clustering, evaluating clusters, profiling segments, and presenting results in a dashboard.

2. Dataset and Preparation

The CSV contains 604 rows before cleaning. It includes client ID, client type, gender, country, region, date of birth, acquisition purpose, loan applied, referral channel, and satisfaction score. Cleaning removed 4 exact duplicate rows and 0 duplicate IDs, leaving 600 unique clients. Age was calculated using 11 November 2026 as the reference date. Missing values were imputed within the modeling pipeline.

3. Methodology

Categorical variables were one-hot encoded; age and satisfaction score were median-imputed and standardized. K-Means candidates k=2 through k=7 were evaluated using inertia and silhouette score. Agglomerative hierarchical clustering was compared at the selected k. The dashboard includes country, region, acquisition purpose, and client type filters.

4. Results (Synthetic Practice Data)

The best tested K-Means silhouette score occurred at k=2, score 0.112. The hierarchical clustering silhouette score at the same k was 0.078. Scores are exploratory and do not prove naturally distinct buyer segments.

K-Means model evaluation

k	Inertia	Silhouette
2	3294.81	0.112
3	3016.23	0.099
4	2887.57	0.088
5	2763.04	0.082
6	2670.38	0.079
7	2613.60	0.076

Cluster profiles

Cluster	Buyers	Median age	Investment	Loan	Satisfaction
0	390	40.0	53.3%	49.5%	4.34
1	210	40.5	59.5%	48.1%	2.79

5. Dashboard Requirements Covered

Buyer segmentation overview, investor behaviour, geographic buyer analysis, segment insights, and filters for country, region, acquisition purpose, and client type.

6. Recommendations

Use cluster profiles to tailor communication; review financing patterns and referral channels on real data; investigate satisfaction differences without assuming causation; validate recommendations using representative data.

7. Limitations and Ethics

The data are synthetic and do not represent actual Parcl customers. Available features do not include property prices, income, budgets, transaction values, or realized returns. Clustering is exploratory and demographic attributes must be handled responsibly.

8. Conclusion

The project demonstrates a reproducible workflow from data cleaning through clustering and dashboarding. Findings must be recalculated on authorized, representative data before company-specific conclusions are made.

Appendix: Reproducibility

Run the notebook from top to bottom. Install dependencies with pip install -r requirements.txt. Launch the dashboard with streamlit run app.py from the project root. Supporting CSV outputs are in the outputs folder.